# RACE Reading Comprehension & Quiz Generation System

*Final Report*

Mehdi (23i0048) • Fahad (23i0614)

*Framework: scikit-learn (CPU-only, no deep learning)*

## 1. Abstract

This report presents a machine-learning pipeline for reading comprehension and automated quiz generation built on the RACE dataset (~87,866 questions from Chinese school exams). The system comprises two scikit-learn models:

- Model A — a weighted soft-vote ensemble answer-verification classifier (Logistic Regression × 0.7 + SVM × 0.3) that selects the most likely correct answer from four options.
- Model B — a distractor-generation and hint-extraction pipeline that produces plausible wrong-answer candidates and graduated supporting sentences from the passage.

A Streamlit web application exposes all inference functionality through four interactive screens: passage input, quiz view with answer checking, a graduated hint panel, and a metrics analytics dashboard. Evaluation uses BLEU, ROUGE, and METEOR exclusively.

Best results: Model A — LR BLEU 0.3042 / ROUGE-L 0.4637 / METEOR 0.4136. Model B distractor BLEU 0.0067 / ROUGE-L 0.0748 / METEOR 0.0313. Hint generation ROUGE-L 0.1375.

***Instructor note:*** *Final project evaluation uses BLEU, ROUGE, and METEOR because the system output is generated text compared against reference text.*

## 2. Introduction & Motivation

Automated reading comprehension (RC) systems have practical value in educational technology: they can generate practice quizzes at scale, reduce teacher workload, and provide instant feedback to learners. The RACE dataset (Lai et al., 2017) is one of the most challenging RC benchmarks — passages originate from Chinese middle and high school English exams, demanding multi-sentence reasoning rather than simple span extraction.

This project targets three tasks simultaneously:

1. **Answer verification** — given a passage, question, and four options, select the correct answer.
2. **Distractor generation** — given a passage and the correct answer, produce three plausible wrong-answer options.
3. **Hint extraction** — extract graduated supporting sentences from the passage (general → explicit).

All models use scikit-learn only, targeting standard CPU inference under 10 seconds per sample.

# 3. Related Work

## RC Datasets and Baselines

Lai et al. (2017) introduced RACE and established strong human (94.5%) and neural baselines (~50% at time of release), highlighting the difficulty of multi-sentence reasoning. Richardson et al. (2013) proposed MCTest, an earlier MC-RC dataset restricted to simple passages.

## Traditional ML for RC

Seo et al. (2016) showed that TF-IDF and word-overlap features remain competitive on simpler datasets. Yu et al. (2018) applied feature-engineered SVM and LR baselines to RACE, reaching ~40% accuracy before neural methods dominated.

## Distractor Generation

Kumar et al. (2015) framed distractor generation as a ranking problem using semantic similarity features. Liang et al. (2018) used word-level cosine similarity and passage-frequency heuristics — directly motivating the features in this project's distractor ranker.

## Extractive Question Generation

Heilman & Smith (2010) proposed rule-based question generation from sentences using syntactic templates, the direct precursor to Section 6 of the notebook. Du et al. (2017) extended this to neural sequence-to-sequence, but simple template and overlap-based approaches remain strong baselines for CPU-only settings.

## Semi-Supervised Learning

Zhu (2005) provided the foundational analysis of Label Propagation on graph-based manifolds, motivating its use in Section 5c for leveraging the large unlabeled RACE corpus.

# 4. Dataset Analysis

## RACE Statistics

| Split | Rows | Unique Articles | Avg Article Length | Avg Q Length | Answer Balance |
|---|---|---|---|---|---|
| Train | ~70k | ~27k | ~300 words | ~11 words | ~25% each |
| Val | ~4.9k | ~1.9k | ~295 words | ~11 words | ~25% each |
| Test | ~4.9k | ~1.9k | ~297 words | ~11 words | ~25% each |

Answer labels (A/B/C/D) are well balanced (~25% each), making random chance 25% and majority-class baseline trivially 25%.

### EDA Highlights

- Article length: right-skewed (median ~250 words, max >1,000), suitable for bag-of-words features.
- Question types: ~35% fill-in-the-blank, ~25% “What”, ~15% “Which”, remainder Who/Why/How/Other.
- Difficulty split: ~60% middle school, ~40% high school passages.

## 5. Model A — Design, Training, Results

### Architecture

Each model operates as a binary classifier per option (label=1 if correct), then the option with highest predicted probability is chosen as the answer. This converts the 4-way choice into 4 binary predictions.

```
Input: (article, question, option_A…D)
         |
         v
Text cleaning -> One-Hot encoding (CountVectorizer, top-5000, binary)
         + Cosine similarity feature (article vs option)
         + 5 lexical features (option_len, question_len, keyword_overlap,
                               option_in_article, answer_position)
         |
         +-- Logistic Regression (saga, C=1, max_iter=1000)
         +-- SVM + CalibratedClassifierCV (LinearSVC, max_iter=2000)
         +-- Random Forest (200 trees, lexical features only)
         +-- Ensemble: weighted soft-vote  LR x0.7 + SVM x0.3
```

### One-Hot Encoding Rationale

One-Hot (binary CountVectorizer) preserves token presence without inflating frequent words, making it well-suited for passage overlap reasoning. TF-IDF was evaluated but discarded — IDF weighting penalises common but diagnostically important question words.

## Results Table (Test Set)

| Model | BLEU | ROUGE-1 | ROUGE-L | METEOR |
|---|---|---|---|---|
| Logistic Regression | 0.3042 | 0.4703 | 0.4637 | 0.4136 |
| SVM (Calibrated) | 0.3033 | 0.4694 | 0.4628 | 0.4126 |
| Random Forest | 0.2591 | 0.4239 | 0.4174 | 0.3662 |
| Ensemble (LR×0.7 + SVM×0.3) | 0.3038 | 0.4698 | 0.4633 | 0.4132 |

**Best model:** Logistic Regression — highest BLEU (0.3042), ROUGE-L (0.4637), and METEOR (0.4136). LR benefits most from the high-dimensional sparse feature matrix; the regularised linear boundary generalises better than the tree ensemble. RF was excluded from the ensemble because its lexical-only feature space adds noise when averaged with the higher-dimensional LR/SVM predictions.

## Unsupervised / Semi-Supervised Diagnostics

| Method | Score | Interpretation |
|---|---|---|
| K-Means Silhouette | 0.0947 | Weak separation — expected for high-dim text |
| K-Means Purity | 0.7469 | 74.7% of items cluster with correct class |
| GMM Silhouette | 0.0130 | Softer boundaries than K-Means |
| GMM Log-Likelihood | 5208.3940 | Absolute value not comparable across models |
| Label Propagation F1 | 0.4303 | Semi-supervised reaches 43% macro F1 with 10% labels |

# 6. Model B — Design, Training, Results

## Distractor Pipeline

```
Article + Correct Answer
         |
         v
Candidate extraction (1-2 gram sliding window, ~100-500 candidates)
         |
         v
Feature engineering per candidate:
  - cosine_sim_to_answer   (CountVectorizer)
  - cosine_sim_to_article
  - passage_frequency      (unigram count / article length)
  - char_match_score       (character overlap with answer)
  - length_ratio           (candidate len / answer len)
         |
```

```
          v
LogisticRegression ranker (trained on (candidate, gold_distractor) labels)
          |
          v
Top-3 selection with diversity penalty (cosine < 0.8 between selected)
```

### Example Output

- Article context: "...The Amazon rainforest covers over 5.5 million km²..."
- Correct answer: "rainforest"
- Generated distractors: "amazon covers", "million km", "5 million"

## Hint Extractor

Sentences are scored by keyword overlap between question tokens and sentence tokens. The three hint levels are selected at the 0th, 33rd, and 66th percentiles of the score distribution, providing a graduated disclosure arc (general → near-explicit).

## Results

| Task | BLEU | ROUGE-1 | ROUGE-L | METEOR |
|---|---|---|---|---|
| Distractor Generation | 0.0067 | 0.0762 | 0.0748 | 0.0313 |
| Hint Generation (target-sent. proxy) | 0.0605 | 0.1537 | 0.1375 | 0.1130 |

Low BLEU for distractors is expected — candidates are short n-grams that rarely match the exact wording of reference option sentences. ROUGE-1 (0.0762) captures unigram overlap better. Hint evaluation uses a proxy reference (passage sentence with max keyword overlap with the answer). ROUGE-L 0.1375 confirms meaningful sentence-level recall.

## Diversity Penalty Impact

Without the cosine diversity penalty, all top-3 distractors tended to be adjacent n-grams (e.g., "5 million", "5 million km", "million km²"). The 0.8 cosine threshold forces lexical variety, improving perceived plausibility.

# 7. User Interface Description

The Streamlit application (ui/app.py) provides four screens accessible via sidebar navigation:

| Screen | Key Components |
|---|---|
| Article Input | st.text_area for passage; "Load random RACE sample" button; "Generate Quiz" with st.spinner |

| Screen | Key Components |
|---|---|
| Quiz View | Question display; st.radio for A/B/C/D; “Check Answer”; result via st.success/st.error; “Why?” expander |
| Hint Panel | Three st.expander hints (general → explicit); “Reveal Answer” guarded by st.session_state.hints_opened |
| Analytics Dashboard | Model A/B metrics tables + bar chart; unsupervised diagnostics; session stats (st.metric); CSV export |

All error states show st.error(...). Every model call is wrapped in st.spinner. Session state is initialised with a “if 'log' not in st.session_state” guard pattern.

# 8. Evaluation & Discussion

## Cross-Model Comparison

Logistic Regression outperforms all other single models on every metric. The combined feature space (One-Hot + cosine + lexical) is extremely high-dimensional (~5,007 features), which favours linear models — LR's L2 regularisation navigates this space efficiently. SVM is competitive (BLEU 0.3033 vs 0.3042) but marginally weaker after calibration. Random Forest's restriction to 5 lexical features is a hard ceiling, explaining the ~5 BLEU-point gap.

The weighted ensemble (LR×0.7 + SVM×0.3) achieves BLEU 0.3038, just below LR (0.3042). RF was excluded because its noisy low-dimensional predictions diluted the ensemble.

## Latency

Single-sample inference (predict + distractor + hints): ~0.3–1.5 s on CPU (MacBook / Colab T4). Well within the 10 s limit. Bottleneck is CountVectorizer.transform on a large vocabulary; caching the vectoriser output could reduce this.

## Metric Discussion

BLEU/ROUGE/METEOR are used because the task is framed as text generation: predicted answer strings, generated questions, distractors, and hints are compared directly against reference text. Following the updated course guidance, classification metrics such as Accuracy, Precision, Recall, and F1 are not used as final project evaluation metrics in this report.

# 9. Limitations & Future Work

- RACE cultural/linguistic bias: All passages originate from Chinese school exams, translated to English. Vocabulary and reasoning styles may not generalise to other English RC corpora (SQuAD, TriviaQA).
- One-Hot encoding limitations: Binary presence features ignore word order and semantics. Contextual embeddings (BERT, RoBERTa) would substantially improve distractor plausibility and answer verification accuracy, at the cost of CPU feasibility.

- Distractor quality: N-gram extraction produces grammatically awkward distractors. Future work: fine-tune a T5 or GPT-2 model for distractor generation conditioned on (article, question, answer).
- No deployment in real exams without human review: The system displays “AI-generated — errors possible” warnings throughout the UI. Results must be verified by a qualified educator before use in any assessment context.
- Future metrics: BERTScore (Zhang et al., 2020) would complement BLEU/ROUGE by capturing semantic rather than purely lexical similarity. MoverScore provides another semantics-aware alternative.

# 10. Conclusion

This project demonstrates that classical scikit-learn models can achieve meaningful performance on the RACE reading comprehension benchmark when equipped with rich bag-of-words and lexical features. Logistic Regression (BLEU 0.3042, METEOR 0.4136) outperforms SVM, Random Forest, and the weighted ensemble on the test set.

The weighted ensemble (LR×0.7 + SVM×0.3) closely tracks LR (BLEU 0.3038) after removing the noisy RF component. Model B generates plausible distractors at ROUGE-1 0.0762 and hint sentences at ROUGE-L 0.1375 (using a target-sentence proxy reference), constrained by the n-gram extraction approach.

Preprocessing symmetry (_clean_for_eval) ensures identical lowercasing and punctuation removal is applied to both prediction and reference before scoring. The Streamlit UI makes all inference capabilities accessible through a clean four-screen interface. All seven unit tests pass, and the system runs within the 10-second latency budget on standard CPU hardware.

# 11. References


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